

RAYAN KHAN

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EDUCATION

Rutgers University

New Brunswick, NJ

B.S. Aerospace Engineering | Expected May 2028 | GPA: 3.54 | Dean's List

Relevant Coursework: Thermodynamics, Fluid Mechanics, Dynamic Systems and Controls

EXPERIENCE

Rutgers University — Low-Reynolds-Number Aerodynamics Lab

New Brunswick, NJ

Undergraduate Researcher

October 2025 – Present

- Built an OpenFOAM/Python CFD pipeline computing lift, drag, and moment coefficients for NACA 0012/2412/4412 airfoils at $Re = 2 \times 10^5$
- Ran 87+ steady RANS simulations from -4° to 20° angle of attack at 0.5° stall resolution; validated results against XFOIL within 4% error
- Engineered a structured C-mesh blockMesh topology that resolved pseudo-3D artifacts; verified pressure and flow fields in ParaView

PROJECTS

Vision-Based "Follow-Me" Drone Controller

June 2026 – July 2026

Python

- Built a real-time "follow-me" drone controller in Python with YOLOv8n tracking and a PID control loop; validated in ArduPilot SITL
- Designed a plugin architecture with abstract source/sink interfaces, decoupling camera, tracking, and control into swappable backends
- Hardened the pipeline against occlusion, calibrated focal length for range estimation, and covered the stack with 32+ unit tests

Two-Body Orbital Propagator with J2 Perturbations

May 2026 – June 2026

Python

- Built a Python orbital-mechanics engine using RK45 integration of the two-body equations of motion; validated J2 regression to 0.4%
- Propagated ISS, GEO, and Molniya orbits from real orbital parameters; generated 3D trajectory plots confirming correct orbital behavior
- Implemented a Hohmann-transfer ΔV calculator and GEO-altitude solver; orbital-element round-trip conversions accurate to $< 1 \times 10^{-8}$

FEA Structural Analysis Tool

March 2026 – May 2026

Python

- Built a linear static finite-element solver from scratch in Python for 3D structural stress, using no commercial FEA package
- Implemented Tet4 and quadratic Tet10 elements with Gauss quadrature; verified correctness via the rigid-body eigenvalue spectrum
- Assembled a sparse global stiffness matrix with direct and conjugate-gradient solvers; applied Dirichlet constraints and consistent nodal loads

Rocket Launch Safety Simulator

February 2026 – May 2026

Arduino, C++

- Led a 6-person team building a closed-loop launch-abort system fusing proximity and temperature sensing to halt unsafe countdowns
- Implemented non-blocking sensor fusion achieving real-time abort detection under 1 second; filtered out servo-induced signal noise
- Architected fail-safe abort logic so that any sensor fault, threshold breach, or timeout defaulted to an immediate launch hold

LEADERSHIP & ACTIVITIES

Accessible 4 All

Co-Founder, Director of Fundraising & Lead Engineer

- Co-founded a registered 501(c)(3) engineering nonprofit that designs and modifies adaptive devices for children with disabilities
- Raised over \$3,500 through donor outreach and internal-affairs operations to fund device builds and materials
- Led teams ranging from 5 to 20+ people to modify toys for children with disabilities, prioritizing strategic delegation

Rutgers University Airborne

General Member

- Conducted a constraint analysis in MATLAB to define mission and flight parameters, establishing a performance baseline for aircraft development
- Modeled mission and airframe components in SOLIDWORKS CAD to GD&T standards for manufacturability
- Authored trade studies and a LaTeX design report submitted to the AIAA Design/Build/Fly competition, justifying key design decisions

SKILLS

Programming & Libraries: Python (NumPy, SciPy, pandas, Matplotlib), C++, C, pytest, OpenCV, YOLOv8, Streamlit, SQLite/SQL, Bash, Git/GitHub

Software & Tools: SolidWorks, Fusion 360, ANSYS Fluent, MATLAB, OpenFOAM, ParaView, XFOIL, gmsh, Arduino, ArduPilot/PX4 SITL

Numerical Methods & Simulation: RK45 integration, finite element method, sparse linear algebra, conjugate-gradient solvers, $k-\omega$ SST RANS, blockMesh C-mesh

Engineering & Analysis: Computational fluid dynamics, FEA, structural analysis, orbital mechanics, control systems (PID), sensor fusion, computer vision, embedded & real-time systems

Languages: English (Native), Urdu (Proficient), Hindi (Limited Working Proficiency)